Chapter 10
Mechanics of Options Markets

Apart from the delivery options embedded in Treasury futures (Section 6.4), everything in Part I was symmetric. A forward, a futures contract, and a swap all oblige both parties, and both can gain or lose without limit. This chapter begins the study of instruments that are not symmetric, and the asymmetry changes almost everything: how they are priced, how they are margined, who can safely write them, and what happens at expiration.

The economics were sketched in Section 1.1. An option confers a right rather than an obligation, so its holder will exercise only when it pays; the payoff can therefore never be negative to the holder, and they must pay for it at the outset. What Part III must supply is the price of that right. First, though, comes the machinery: what exactly the contract says, how it is quoted, who guarantees it, what the writer must post, and what happens at expiration. Getting this wrong is a common source of loss, and unlike model error it is avoidable.

Sections 10.1 to 10.4 are the contract and how it is read: the four positions, the specifications, the underlyings beyond single stocks, and the option chain. Sections 10.5 and 10.6 are how options trade and are quoted, including the volatility quotation Chapter 15 explains. Sections 10.7 to 10.9 are the mechanics of short selling, margin, and exercise and assignment; Section 10.10 looks beyond the standard contract and Section 10.11 at who trades options and why.

This chapter also covers a development with no counterpart in earlier textbooks: options markets have been transformed since 2020 by retail participation and by contracts that expire the same day they are traded. Section 10.12 treats it as a market-structure topic rather than a curiosity, because it now shapes intraday behavior in the largest equity index in the world.

10.1 CALLS, PUTS, AND THE FOUR POSITIONS

A **call** gives its holder the right to buy the underlying asset at the **strike price** K; a **put** gives the right to sell at K. The right lasts until the **expiration date**. A **European** option can be exercised only at expiration; an **American** option can be exercised at any time up to it. The names, coined by Paul Samuelson in the mid-1960s, have nothing to do with geography: both types trade on both continents.

Every option contract has two sides. The holder is said to be **long** and has **bought** the option; the writer is **short** and has **sold** it. There are four elementary positions, shown in Figure 10.1. Like Figure 1.1, the figure plots profit, the payoff net of the premium, because that is what a trader experiences.

The two right-hand panels are the ones to dwell on. A writer's best case is the premium, achieved when the option expires worthless, and their worst case is unbounded for a call and bounded only by the stock reaching zero for a put. This is why the two sides of an option trade are not mirror images in any practical sense: they face different margin requirements, different capital treatment, and different credit risk. The buyer of an option has no obligations after paying the premium and can be a retail client with a small account. The writer has an open-ended obligation and must be collateralized, which is the subject of Section 10.8.

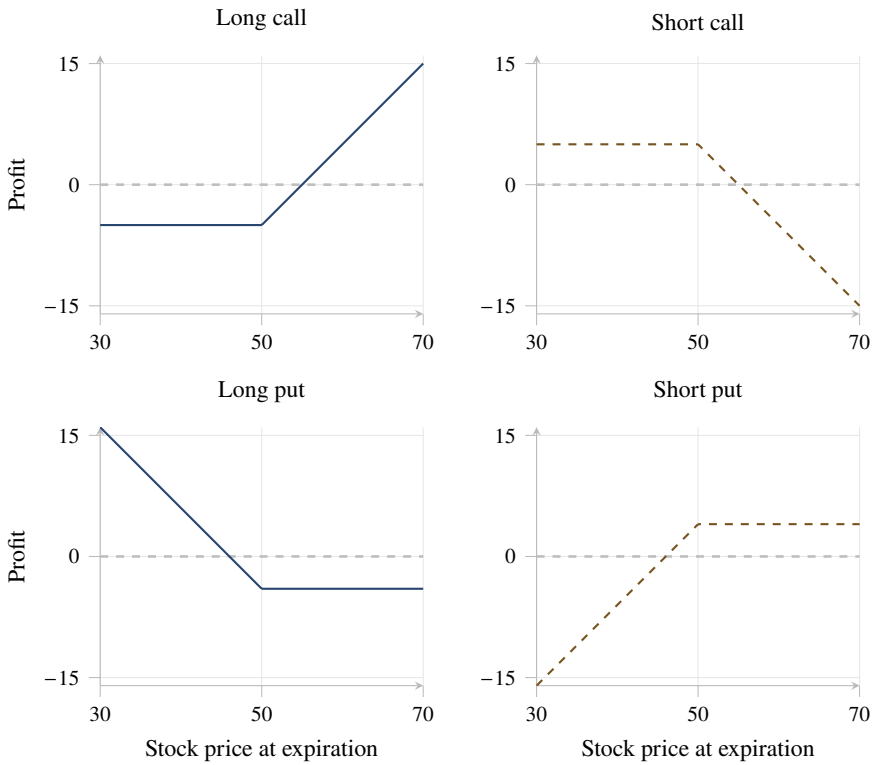


Figure 10.1: Profit at expiration. Strike 50; call premium 5, put premium 4.

Moneyness. An option's **moneyness** is where the stock price S stands relative to the strike K. An option is **in the money** if exercising it immediately would produce a positive cash flow, **at the money** if the stock price equals the strike, and **out of the money** otherwise. For a call these correspond to $S > K$, $S = K$, and $S < K$; for a put the inequalities reverse. The terms describe the option, not the trade: an out-of-the-money option can be an excellent purchase and an in-the-money one a poor purchase, depending entirely on price.

Intrinsic and time value. The **intrinsic value** of a call is $\max(S - K, 0)$, and of a put $\max(K - S, 0)$: what the option would be worth if it expired now. An American option must trade for at least its intrinsic value, since otherwise one could buy it and exercise immediately for a riskless profit. The difference between the market price and the intrinsic value is the **time value**, and it exists because the remaining life gives the stock further opportunity to move in the holder's favor while the downside stays capped.

Figure 10.2 shows the relationship that Part III spends several chapters making precise. Suppose the stock is at 50 and a call struck at 50 trades at 5. Its intrinsic value is zero, so all 5 is time value. With the stock at 60 the curve puts the call at about 12: 10 of intrinsic value and about 2 of time value. With the stock at 40 it is worth about 2, again all time value. Two features generalize. Time value is greatest near the money, because that is where uncertainty about whether the option finishes in or out of the money is greatest; deep in-the-money and out-of-the-money options have little of it. And time value decays to zero at expiration, when the curve collapses onto the kink, the phenomenon Chapter 18 quantifies as theta.

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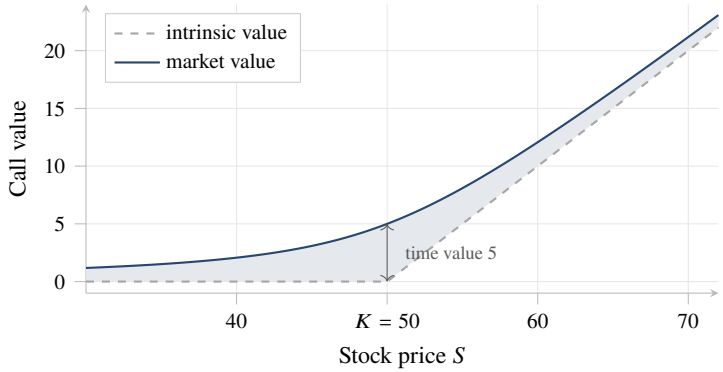


Figure 10.2: A call before expiration, assumed to trade at 5 with the stock at the strike of 50. Shaded region is time value. Schematic.

10.2 CONTRACT SPECIFICATIONS

An exchange-traded option is a standardized contract, and the standard terms matter because a trader chooses among them rather than negotiating them.

Contract size. A standard US equity option covers 100 shares. A quoted premium of $3.20 therefore costs $320 per contract, and a novice who confuses the two has misjudged their position by two orders of magnitude.

Strike prices. The exchange lists strikes at fixed intervals around the current price and adds new ones as the stock moves. The base rule for US equity options sets the interval by price level: $2.50 for strikes up to $25, $5 from $25 to $200, and $10 above $200. Listing programs have narrowed that considerably. The $2.50 Strike Price Program extends the $2.50 interval on selected classes to $50, and to $100 within $10 of the stock price; most actively traded stocks carry $1 intervals below $50, and the lowest-priced stocks $0.50. The largest ETFs and indices are denser still: SPY, the largest S&P 500 ETF, is listed at $1 intervals whatever its price, and S&P 500 index options at 5-point intervals near the money, widening to 25 and 50 points farther out. An actively traded $48 stock therefore shows strikes at 45, 46, 47, and so on up to 50, then 52.50, 55, 57.50, and 60. New strikes are added as the stock nears either end of the ladder, so the price stays bracketed, and old ones stay listed until they expire. This is why a stock that has doubled carries a long tail of near-worthless low-strike puts beneath it.

Expiration dates. US equity options historically expired monthly, on the third Friday. That is still the anchor for the largest contracts, but the calendar has filled in dramatically. Weekly expirations were introduced in 2005 and extended across the most liquid stocks, and since the week of January 26, 2026, after SEC approval that month, stocks with a market value above $700 billion and the largest bitcoin ETF have also carried Monday and Wednesday expirations. For the S&P 500 index, expirations now occur *every trading day*, following the addition of Tuesday and Thursday expirations in 2022. Figure 10.3 shows what this looks like over a typical month, and Section 10.12 explains why it matters.

Exercise style. Options on individual US stocks are American; options on broad indices are typically European. The distinction is not cosmetic. Chapter 11 shows that an American call on a non-dividend-paying stock should never be exercised early, that an American put may well be, and that the difference in value between the two styles can be substantial for puts.

Settlement. Equity options deliver shares. Index options settle in cash against a value computed from opening or closing prices, because delivering a 500-stock basket is impractical.

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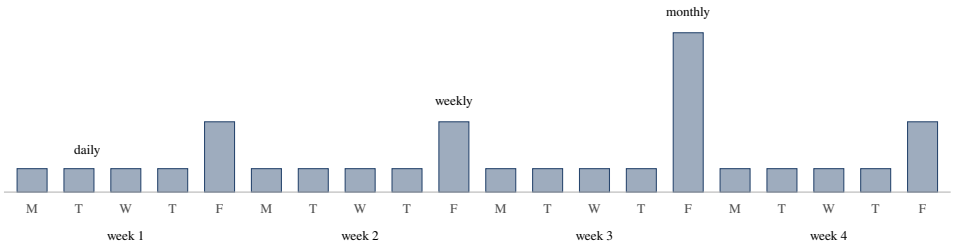


Figure 10.3: S&P 500 index option expirations in a month. Bar heights schematic.

Corporate actions and contract adjustment

An option is a claim on a specified quantity of a specified security. When a corporate action changes what a share *is*, the contract must be adjusted, or the option would be arbitrarily enriched or destroyed by an event unrelated to the company's value.

The governing principle is to preserve the aggregate value of the position. For a stock split or stock dividend, the contract multiplier and strike are adjusted so that the total strike cost is unchanged. For a merger, the deliverable becomes whatever a holder of 100 shares receives. Ordinary cash dividends, paid under a regular policy, do *not* trigger an adjustment, however large; a special dividend of \$0.125 a share or more does, normally by reducing the strike. The exclusion of ordinary dividends is often misunderstood, and Chapter 11 shows that it is why dividends can make early exercise of American calls rational.

Example 10.1 A three-for-two split

An investor holds 10 call contracts on a stock, strike \$60, each covering 100 shares. The company declares a 3-for-2 stock split; each old share becomes 1.5 new shares, and the stock price falls from about \$66 to about \$44.

Without adjustment the options would be destroyed: a \$60 strike on a \$44 stock is deep out of the money, though nothing of economic substance has happened. The adjustment multiplies the deliverable by $3/2$ and divides the strike by the same factor:

$$\text{new contract size} = 100 \times \frac{3}{2} = 150 \text{ shares}, \qquad \text{new strike} = \frac{60}{3/2} = \$40$$

Check that the position is unchanged. Before, exercising cost $10 \times 100 \times 60 = \$60{,}000$ for 1,000 shares worth \$66,000. After, exercising costs $10 \times 150 \times 40 = \$60{,}000$ for 1,500 shares worth $1{,}500 \times 44 = \$66{,}000$. The strike cost, the share value, and the moneyness are all preserved.

The number of contracts stayed at 10 and the contract size changed. For splits producing whole-number ratios (2-for-1, 3-for-1) the convention runs the other way: the contract count is multiplied and the size stays at 100. Either way the aggregate is preserved, but a trader reading a position report must know which convention was applied.

How an option is named

Every listed option needs an identifier, and unlike a stock ticker it must encode four things at once: the underlying, the expiration, the type, and the strike. Since 2010 the US market has used the format established by the Options Symbology Initiative, shown in Figure 10.4.

The symbol in the figure reads: Apple, expiring January 19, 2024, call, strike \$190.000. Nothing has to be looked up: the identifier is the contract description.

The strike is carried in eight digits with three implied decimals, so `00190000` is \$190.000 and `00002500` is \$2.50. The three decimal places exist precisely because adjustments of the kind in Example 10.1 can produce strikes that are not round numbers: dividing \$60 by $3/2$

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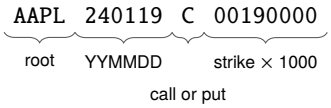


Figure 10.4: US option symbol format.

gave a clean \$40, but dividing by $7/5$ would not have.

This scheme replaced a much cruder one. Before 2010, an option was identified by a root plus two letters, one encoding the expiration month and type and one encoding the strike. With only twenty-six letters available for strikes, the system could not survive the proliferation of strike prices and expirations described earlier in this section, and by the late 2000s it was producing genuine ambiguity about which contract a symbol referred to.

The adjusted-contract trap. One convention deserves emphasis because it causes real losses. When a contract is adjusted for a corporate action, it does not keep the ordinary root but receives a *numbered* one: `AAPL1`, then `AAPL2` for a second adjustment, and so on. The digit is appended whatever the length of the root; the root field of the symbol is six characters wide, so even a five-letter root such as `GOOGL` has room for it. The two roots then trade side by side, and they are different contracts. `AAPL` delivers 100 shares; `AAPL1` may deliver 150 shares, or shares plus cash, or shares in an acquiring company.

A trader who sees an unusually cheap quote on what appears to be a familiar option is very often looking at an adjusted contract with a different deliverable, and a position report that lists both roots under the same underlying will misstate the exposure. The rule is simple: a numbered root means read the adjustment memorandum before trading.

10.3 OPTIONS ON INDICES, ETFS, CURRENCIES, AND FUTURES

Options on individual shares are the natural teaching case but not the largest market. Table 10.1 sets out how the specifications differ across underlyings, because the differences change what a position actually is.

Table 10.1: Specifications across underlyings, US market.

Underlying	Size	Exercise	Settlement	Note
Single stock	100 shares	American	Deliver shares	Dividends drive early exercise
Stock index	Multiplier × index	European	Cash	Settlement value may be untradeable
Index ETF	100 shares	American	Deliver shares	Index economics, stock mechanics
Currency	Fixed currency amount	European	Cash or delivery	Foreign rate acts like a dividend yield
Futures	One futures contract	American	Futures position	Exercise yields a margined position

Index options. A stock index option has a multiplier (\$100 for the main US index contracts), so a contract on an index at 5,250 controls $5{,}250 \times 100 = \$525{,}000$ of notional. These are the largest option contracts most participants will encounter. Example 10.2 works through a position.

Cash settlement introduces a subtlety. Monthly index options settle against a value computed from the *opening* prices of the constituent stocks, which do not all open at once. The resulting settlement value can differ from any index level published during the session, and a holder

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cannot trade at it. A position carried into settlement is therefore exposed to a price that is calculated rather than observed.

Example 10.2 An index option position

An index stands at 5,250. A fund buys 10 call contracts struck at 5,300, expiring in one month, quoted at 42.00. The multiplier is \$100.

The position costs $10 \times 100 \times 42.00 = \$42{,}000$ and controls $10 \times 100 \times 5{,}250 = \$5{,}250{,}000$ of index. The fund has obtained exposure to \$5.25 million of index for \$42,000 (eight-tenths of one percent) and, unlike the futures position of Example 1.5, it can never be asked for another cent. That is the main argument for using options when funding is uncertain.

If the index settles at 5,380, the payoff is $(5{,}380 - 5{,}300) \times 100 \times 10 = \$80{,}000$ and the profit is \$38,000. If it settles anywhere at or below 5,300 the options expire worthless and \$42,000 is lost. The index must rise about 1% merely to reach the strike, and about 1.8% to break even: buying out-of-the-money options requires being right about direction, size, and timing simultaneously, which is why Chapter 12 builds structures that relax one of those requirements.

ETF options. Options on an index-tracking ETF give near-identical economic exposure but with the mechanics of a stock option: American exercise, delivery of ETF shares, and dividends paid by the fund. Since November 2024 they have included the spot bitcoin ETFs, the largest now with a position limit of 1,000,000 contracts. A trader who is short ETF calls can be assigned early, that is, made to deliver because a holder has exercised (Section 10.9). The risk peaks around an **ex-dividend date**, the first day the shares trade without the coming dividend, because a holder can exercise the day before to collect the dividend. A trader short European index calls such as S&P 500 index options (SPX) cannot be assigned at all before expiration. (The S&P 100 trades both ways: OEX, the first index option, listed by the Chicago Board Options Exchange (CBOE) in 1983, is American, and XEO is its European twin.) The two positions look interchangeable on a payoff diagram and are not interchangeable in a risk system.

There is also a tax distinction, and in the US it is large enough to drive the choice between the two. Broad-based index options are "Section 1256" contracts: every gain or loss is split, with 60% taxed at the long-term capital gains rate and 40% at the short-term rate, however briefly the position was held, and open positions are marked to market at year end. Options on an index ETF are ordinary equity options and receive no such treatment: a position held for a month is taxed entirely at the short-term rate. For a taxable US investor trading short-dated index exposure, this is often a larger consideration than the bid–ask spread.

Currency options. A currency option is written on a fixed amount of foreign currency, and its valuation treats the foreign interest rate exactly as a dividend yield, the reasoning of Section 5.7 carried into option pricing. Chapter 16 develops it.

Options on futures. Exercising an option on a futures contract does not deliver an asset; it delivers a *position* in the futures contract, marked to market from the start. Exercising a futures call therefore produces a long futures position at the most recent settlement price plus a cash payment of that settlement price less K; a holder who closes the position at once collects the intrinsic value at the futures price then prevailing. Because the resulting position is itself margined, an option on futures never requires the holder to fund the underlying, which is why these contracts dominate exchange-traded options on commodities and interest rates. Producers and processors use them to set a floor or a ceiling on what they sell or buy, and banks and asset managers to hedge rate moves. Chapter 16 prices them with Black's model, and Chapter 6's SOFR and Treasury contracts are the underlyings that matter most.

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10.4 READING AN OPTION CHAIN

Options are quoted in a grid (strikes down, expirations across) known as the **option chain**. Table 10.2 shows one expiration for a stock at $48.30. The chain is shown at $2.50 intervals for compactness; an active stock at this price would also carry the $1 strikes between them.

Table 10.2: Calls and puts on a stock at $48.30, 45 days to expiration. Vol is volume and OI open interest.

Calls				Strike	Puts			
Bid	Ask	Vol	OI		Bid	Ask	Vol	OI
6.55	6.75	120	3,104	42.50	1.55	1.62	890	9,220
4.35	4.50	410	5,870	45.00	1.81	1.89	2,140	14,505
2.62	2.72	3,880	21,440	47.50	2.52	2.63	3,410	18,760
1.44	1.51	5,260	28,115	50.00	3.80	3.95	1,505	11,230
0.68	0.74	2,940	19,680	52.50	5.55	5.75	240	4,010
0.28	0.33	1,110	12,450	55.00	7.60	7.90	60	1,845

Example 10.3 What the chain is telling you

Read Table 10.2 carefully and several facts emerge that no formula supplies.

Where the trading is. Volume and open interest peak at the 50 and 47.50 strikes, just above and below the $48.30 spot. Liquidity concentrates near the money, and a trader wanting the 55-strike call must accept a bid–ask spread that is wider in proportion to the premium.

The cost of the spread. The 50-strike call is 1.44 bid, 1.51 ask. A trader who buys and immediately sells loses $0.07 per share, or $7 per contract, or 4.7% of the position's value. On the 55-strike call the spread is $0.05 on a $0.30 option, which is over 16%. Bid–ask spreads on cheap options are enormous in percentage terms, and a strategy that trades far out-of-the-money options frequently can be defeated by transaction costs alone.

Volume versus open interest. Volume is contracts traded today; open interest is contracts outstanding. At the 42.50 put, volume of 890 against open interest of 9,220 says most holders are sitting still. High volume relative to open interest (the 47.50 call, 3,880 against 21,440) indicates active repositioning.

An internal check. Put–call parity, derived in Chapter 11, requires that a call and put at the same strike be linked through the stock price and the discounted strike. At the 47.50 strike, the call's midpoint, 2.67, and the put's, 2.575, differ by $0.095. With no dividends and a 5% rate over 45 days, the difference would have to be

$$S - Ke^{-rT} = 48.30 - 47.50\,e^{-0.05\times 45/365} = 48.30 - 47.208 = \$1.092$$

The shortfall of about $1.00 is mostly the market's expectation of dividends payable before expiration, together with any cost of borrowing the stock, which the quotes alone cannot separate from them. Every strike from 42.50 to 50 implies about the same $1.00 (the two highest read a few cents more, for a reason Chapter 11 gives), so the chain is internally consistent, and Section 11.6 shows how to read the number off the quotes.

10.5 TRADING AND MARKET MAKING

US listed options trade on more than a dozen exchanges, all clearing to the same clearing house, with orders routed among them by price and by economics that the end investor does not see. The SEC regulates them and the Financial Industry Regulatory Authority (FINRA) the brokers that sell them; options on futures are futures products, regulated by the CFTC. Two features of this structure have direct consequences.

Market makers quote a bid and an ask continuously and are the counterparty to most

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retail trades. They do not take directional views; they earn the spread and hedge the resulting exposure, dynamically, using the underlying and other options. Chapter 18 is about how that hedging works. What matters here is the consequence: a market maker who has sold calls is short the option and must buy stock as the stock rises, which means dealer hedging *transmits* option activity into the underlying market. Section 10.12 shows what happens when the option activity becomes large enough relative to the stock.

Payment for order flow. Retail brokers in the US route customer orders to wholesalers (large market makers that fill retail orders), which pay for the privilege, and since 2019 this has funded zero-commission trading. The trade is not free; the cost has moved from an explicit commission into the execution price, and it is the harder cost to see. Whether retail investors are better off on net is contested: execution quality improved for stocks, but the picture is murkier for options, where spreads are wider and **price improvement** (the amount by which a fill beats the best displayed quote) is measured against a quote so wide that beating it says little.

Position limits, order types, and legging risk

Three further pieces of trading mechanics have practical consequences.

Position and exercise limits. Exchanges cap the number of contracts any one participant may hold on the same side of the market (calls held plus puts written, or puts held plus calls written), with limits ranging from tens of thousands of contracts on less actively traded stocks to 250,000 on the most liquid stocks, and millions on the largest ETFs. Exercise limits cap how many may be exercised within any five consecutive business days, generally at the same number as the position limit. The purpose is to prevent the option market being used to corner the underlying: a call position large relative to a stock's free float (the shares actually available for trading), exercised at once, can force short sellers to buy back shares at inflated prices. Bona fide hedgers can seek exemptions, as from the futures limits of Section 2.1.

Order types. Because option bid–ask spreads are wide relative to the premium (Example 10.3 showed 4.7% and 16%), a market order is genuinely dangerous. It executes at the ask when buying and the bid when selling, on a quote that may itself have widened in the instant before it arrives. Professional practice is to place limit orders at or near the midpoint and accept non-execution; a retail trader who submits market orders in options pays the full spread on every round trip.

Legging risk. A strategy with two or more option legs can be executed leg by leg or as a package. Executing separately exposes the trader to the market moving between fills, which on a two-option position worth $0.40, built from options worth $2.60 and $2.20, can easily exceed the value of the position itself. Exchanges therefore maintain complex order books that match multi-leg orders as a single package at a net price. Chapter 12 builds strategies from several legs throughout, and every one of them should be understood as a package rather than as a sequence of trades.

10.6 QUOTING IN VOLATILITY

A feature of professional option markets surprises newcomers: in large negotiated trades, dealers do not negotiate the price. They negotiate the implied volatility, which Section 15.7 defines as the volatility that, put into the pricing formula, returns the option's price. The price follows once the trade is agreed.

The reason is practical. An option's price depends on the underlying price, the strike, the time remaining, interest rates, and volatility. Four of those five are observed in the market or fixed by the contract; only volatility is not, and it is where the two sides' views differ. If a dealer quotes "$1.47" and the stock moves a penny before the client responds, the quote is stale. If the dealer quotes "21.5 volatility," the quote survives the stock moving, and the price is struck at

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the end from the agreed volatility and an agreed reference spot. Two consequences follow, and both matter for Part III.

First, large OTC option trades are commonly done with a **delta exchange**: the option trades together with an agreed quantity of the underlying, sized to the option's directional exposure (its delta, in the language of Chapter 18), so that neither side takes on a position in the stock they did not want. The trade is a pure volatility transaction, which is what both sides intended.

Second, and more fundamentally: implied volatility is a price, not a model output. It is tempting to regard it as something extracted from an option price by inverting a formula. In that market the causation runs the other way: volatility is quoted, negotiated, and traded, and the dollar price is derived from it. That is why Chapter 19 studies the resulting surface of quoted volatilities across strikes and maturities as a market object in its own right.

10.7 SHORT SELLING, BORROW COSTS, AND PUTS

One more piece of mechanics connects the option market to Part I, and it explains a pattern that otherwise looks like mispricing. Buying a put and writing a call at the same strike and expiration creates a position that gains and loses exactly as a short stock position does, a **synthetic short**. Chapter 11 derives this from put–call parity. It means an investor who wants short exposure has two routes: borrow the stock and sell it, or build the synthetic in the option market.

The two routes must cost about the same, which has a direct consequence. Section 5.2 showed that when a stock is hard to borrow and its holders will not or cannot sell it, the no-arbitrage relation degrades from an equality into a band whose width is set by the borrow fee. That band appears in the option market too: on a hard-to-borrow stock, puts trade above the price that put–call parity with the calls implies, by roughly the present value of the borrow cost. Example 10.4 puts a number on it.

Example 10.4 What a borrow fee does to the option chain

A stock trades at \$50 and can be borrowed only at a fee of 3% a year. Consider six-month options struck at \$50.

An arbitrageur who saw the put priced above what put–call parity implies would sell it, buy the call, and short the stock (the reversal of Chapter 11), but shorting requires borrowing at 3%, which over six months costs

$$50\left(1 - e^{-0.03\times 0.5}\right) = \$0.74$$

per share. The trade is therefore unprofitable unless the put is more than 74 cents rich, and the option market can sit anywhere inside that band without offering a profit to anyone who must borrow the stock. (A holder willing to sell shares already owned avoids the fee, which is why the band appears only where such holders are scarce, as they usually are on a heavily shorted stock.)

So a chain showing puts 60 cents rich to naive put–call parity is not mispriced and does not indicate bearish sentiment. It is quoting the borrow fee. A trader who reads it as a signal has read a financing cost as an opinion. Chapter 19 finds something similar in the volatility skew, the pattern in which low-strike options trade at higher implied volatilities than high-strike ones: part of it is a price set by one-sided demand for protection rather than a forecast.

The magnitude scales with the fee, and for genuinely hard-to-borrow stocks the fee can reach tens of percent, at which point the put becomes the only practical way to be short and its price says almost nothing about the expected direction of the stock.

10.8 MARGIN REQUIREMENTS

The asymmetry of Figure 10.1 produces an asymmetry in collateral, and it is the cleanest illustration in the book of margin following risk rather than notional.

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The buyer posts nothing. A purchased option is paid for in full at inception (US rules let a listed option with more than nine months to run be bought with 75% paid and the rest borrowed) and creates no further obligation. (The exception is an option margined futures-style, Section 16.5, whose value change is settled daily.) The maximum loss is the premium, already paid. This is what makes options valuable to a hedger with limited access to liquidity (Section 12.10).

The writer posts margin. A **naked** call writer, one who has sold calls without owning the shares, has unlimited exposure and must collateralize it. The standard US requirement is the greater of two formulas:

$$\text{(a)}\quad 20\%\text{ of underlying value} - \text{out-of-the-money amount} + \text{option value} \tag{10.1}$$

$$\text{(b)}\quad 10\%\text{ of underlying value} + \text{option value} \tag{10.2}$$

The option value is the premium on the day of the trade and the market price thereafter, so the requirement is recomputed daily as the option moves. The second formula is a floor that binds for far out-of-the-money options, where the first would permit a very large position on very little collateral. A naked put writer posts the greater of the same two formulas, with the out-of-the-money amount measured as how far the stock stands above the strike and the floor in (b) taken as 10% of the strike price rather than of the underlying value. Example 10.5 applies both to a call.

Example 10.5 Margin on a naked call

A trader writes 10 naked call contracts, strike \$55, on a stock at \$48.30, collecting \$0.30, about the midpoint of the 0.28–0.33 quote in Table 10.2. The premium received is $10 \times 100 \times 0.30 = \$300$.

Underlying value is $10 \times 100 \times 48.30 = \$48{,}300$, and the options are \$6.70 out of the money, or $10 \times 100 \times 6.70 = \$6{,}700$ in aggregate. From (10.1),

$$0.20 \times 48{,}300 - 6{,}700 + 300 = 9{,}660 - 6{,}700 + 300 = \$3{,}260$$

and from (10.2),

$$0.10 \times 48{,}300 + 300 = \$5{,}130$$

The requirement is the greater, \$5,130.

The trader has collected \$300 and must post \$5,130, seventeen times the premium, to support a position whose maximum gain is that \$300 and whose maximum loss is unbounded. If the stock rallies to \$70, the calls are worth at least \$15 and the position has lost at least $10 \times 100 \times 15 - 300 = \$14{,}700$, nearly three times the margin originally posted. As in Example 1.5, margin is a deposit and not a maximum loss, and the writer will be called for the difference.

This is the arithmetic behind a standing piece of market wisdom: selling far out-of-the-money options produces a long run of small gains punctuated by losses that erase them. The strategy requires capital, discipline, and the ability to survive the tail. It is not a yield-enhancement technique, whatever it is marketed as.

Covered positions require less. A call writer who owns the underlying shares has no unbounded exposure (the shares are the deliverable) and posts no additional margin beyond the stock itself. A cash-secured put writer who holds K per share in cash is similarly covered. These structures dominate retail option writing for exactly this reason.

Portfolio margining evaluates the whole position rather than each leg, stressing the portfolio across a grid of underlying prices and volatilities and requiring collateral for the worst outcome. The methodology is the option-market cousin of the portfolio margining of Section 2.3 and of the initial margin models of Chapter 8. It substantially reduces requirements for genuinely hedged positions and, as Section 5.14 illustrated with Amaranth, thereby permits positions whose size becomes the risk.

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10.9 THE CLEARING HOUSE, EXERCISE, AND ASSIGNMENT

Every US listed option is cleared by the Options Clearing Corporation (OCC), which was established in 1973 alongside the CBOE and remains the sole clearing house for the market. The OCC becomes buyer to every seller and seller to every buyer, so an option holder never depends on the solvency of the person who wrote it. The mechanism and the risk it concentrates are those of Section 2.4.

Exercise. Most positions are never exercised but closed by an offsetting trade, the holder selling the option and the writer buying it back, which also captures any remaining time value. A holder exercising an American option notifies their broker, who notifies the OCC. At expiration, options that are in the money by one cent or more are exercised *automatically* ("exercise by exception") unless the holder instructs otherwise. This default prevents value being abandoned through inattention, but it does not apply to an option that finishes exactly at the strike, the trap of Example 10.6.

Assignment. The OCC does not know which particular writer should be assigned, and it does not try to. It allocates exercise notices *randomly* among clearing members with short positions, who in turn allocate randomly or first-in-first-out among their clients. A writer therefore cannot know in advance whether they will be assigned, only estimate the probability. A writer of an in-the-money American option can have it exercised against them at any moment.

Example 10.6 Pin risk at expiration

A market maker is short 200 put contracts struck at \$50 and has hedged by shorting 10,000 shares, roughly the hedge Chapter 18 will call the delta hedge: with the stock near \$50 the puts behave like about half of their 20,000 shares. As expiration approaches, though, the puts are almost certain to be exercised if the stock finishes below \$50 and certain to expire worthless if it finishes above.

The stock closes at exactly \$50.00 on expiration Friday. If every holder exercises, the writer is assigned 20,000 shares at \$50, which covers the 10,000-share short and leaves 10,000 shares to sell. If no one exercises, the writer is left short 10,000 shares they did not want. In practice some holders exercise and some do not, for reasons the writer cannot observe, and the writer learns the outcome only after the market has closed: on Saturday morning, with the next opportunity to trade on Monday.

The exposure is therefore an unhedged overnight-plus-weekend position of unknown size and unknown direction, on up to 10,000 shares. This is **pin risk**, and it is why traders close positions at strikes the stock is likely to finish near, rather than carrying them into expiration for the last few cents of premium.

A related effect runs the other way. A dealer who is *long* options struck near the spot (long gamma, in Chapter 18's terms) hedges by buying stock when it falls below the strike and selling when it rises above, which mechanically damps movement around that price. Stocks with large open interest at a round strike tend to gravitate toward it as expiration approaches. This "pinning" effect is a hedging artifact rather than a market view. The writer in this example, being short, would trade the other way, which is why pinning depends on who holds the options at the strike.

10.10 BEYOND THE STANDARD CONTRACT

Three families of option-like instruments sit outside the listed market and are worth distinguishing, because their differences from a listed option are exactly the features a reader should be alert to.

Warrants are options issued by a company on its own stock, usually attached to a bond or share issue so that it raises capital more cheaply, while holders gain long-dated upside. Exercise creates new shares, so the payoff is diluted, which changes the valuation (Chapter 17). A listed call, by contrast, is written by another participant and exercise transfers existing shares.

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Convertible bonds are debt with an embedded option to convert into equity. The holder owns a bond plus a call; the issuer has sold that call in exchange for a lower coupon. They form a large market in embedded options, and Chapter 25 treats them together with credit.

Employee stock options are non-transferable, forfeited if unvested when the employee leaves, and long-dated. Because holders cannot sell them, they exercise early for liquidity and diversification reasons that no arbitrage-based model captures, which is why the valuation of employee options is a distinct problem, the subject of Chapter 17.

Alongside them sits the over-the-counter market in options proper. These are options in the full sense of Section 10.1, negotiated bilaterally with a dealer rather than listed. The listed market is the visible one; by notional the OTC market is larger, and it is where most currency options, most interest rate options (the caps, floors, and swaptions of Chapter 29), and every exotic structure of Chapter 34 trade. The reason is customization. An exchange lists a grid of strikes and dates on a hundred shares of one stock; a dealer will write any strike and any date, on a basket, with a barrier, on an average price, or paid in another currency, in amounts that would visibly move a listed market. The terms are documented under the ISDA Master Agreement of Section 1.5, the price is negotiated as a volatility in the manner of Section 10.6, and the buyer is exposed to the dealer's default for the life of the trade unless the agreement is collateralized under a CSA of the kind Chapter 8 describes or the contract is standardized enough to clear. The credit valuation adjustment of Chapter 9 is the price of that exposure. The users are the ones the listed market cannot serve: a treasurer hedging a receivable whose date and amount match no listed contract, an asset manager buying a swaption against a pension fund's interest rate exposure, an issuer of the structured notes of Chapter 35 hedging the option it has sold to retail investors. What the buyer gives up is the exchange's transparency: there is no public last trade, the dealer's quote is the only price, and a position is unwound with the same dealer or not at all.

Flexible exchange (FLEX) options, introduced in 1993, are the listed market's answer: exchange-listed, but with terms the parties choose, namely any strike, any expiration, either exercise style. They occupy the middle ground the cleared-OTC market occupies in Section 1.4, offering customization with a clearing house behind it. **Long-term equity anticipation securities (LEAPS)** are simply long-dated listed options, extending up to about three years (five on the S&P 500).

10.11 WHO TRADES OPTIONS, AND WHY

Options serve purposes the linear instruments of Part I cannot, and it is worth naming who is on each side before Part III begins pricing them.

Protective put buyers own an asset and want a floor under it. The put is insurance: a premium is paid, the downside is capped, the upside is retained. Pension funds, insurers, and founders and executives with concentrated stock positions are the natural buyers, and Section 12.10 explains why a futures hedge is not a substitute.

Covered call writers own an asset and sell part of the upside for income. Asset managers run this as a systematic strategy, and it is among the most widely used retail option strategies. It converts an uncertain equity return into a more certain but capped one, and it performs poorly in the strong markets that investors hold equities for, as Example 10.7 shows.

Example 10.7 Writing a covered call

An investor owns 500 shares of the stock in Table 10.2, trading at \$48.30, and writes 5 call contracts struck at \$52.50 at the \$0.68 bid, 45 days out. Premium received is $5 \times 100 \times 0.68 = \$340$.

Three outcomes at expiration follow, on the 500-share position:

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Stock at $46. The calls expire worthless. The shares have lost $500 \times 2.30 = \$1{,}150$, offset by the \$340 premium, for a net loss of \$810 against \$1,150 unhedged.

Stock at $52.50. The calls expire just at the money and worthless. The shares have gained $500 \times 4.20 = \$2{,}100$, plus \$340, for \$2,440. This is the best possible outcome, and every price at or above the strike earns it.

Stock at $58. The calls are assigned at \$52.50. The result is the same \$2,440, while the unhedged position would have gained $500 \times 9.70 = \$4{,}850$. The investor gave up \$2,410.

The premium is \$0.68 on a \$48.30 stock: 1.4% of downside protection, and the breakeven falls only to \$47.62. The strike is 8.7% above spot, so all upside beyond 8.7% has been sold for that 1.4%.

The maximum return is \$2,440 on \$24,150 of stock, or 10.1% in 45 days, which annualizes to a figure prominent in marketing material. Annualizing it is misleading: the maximum cannot be earned repeatedly, because it requires the stock to gain 8.7% or more every 45 days.

The strategy is not unsound: the writer is selling insurance and collecting a premium for it, and Chapter 19 shows that on average the premium exceeds what the options pay out. But "income" misdescribes the strategy: it is the sale of upside.

Corporate issuers embed options in securities. A callable bond is a bond plus a call sold to the issuer, who pays for the right to refinance with a higher coupon; a convertible is the reverse, a call sold to the holder in exchange for a lower coupon.

Volatility traders take positions in options without wanting exposure to the underlying at all, hedging the direction away and trading the volatility that remains. Part IV is about this.

Dealers intermediate all of the above and hedge what accumulates. Their aggregate position determines whether dealer hedging damps or amplifies moves in the underlying, the mechanism of Example 10.6 operating at market scale.

Retail investors have become, since 2020, a large enough share of activity to change market structure, which is the subject of the next section.

10.12 THE RETAIL TRANSFORMATION AND SAME-DAY OPTIONS

The US options market more than doubled in volume between 2019 and 2023, and the composition of that growth is as important as its size. Three changes drove it.

The first was the elimination of retail commissions in late 2019, which removed the fixed cost that had made small option trades uneconomic. The second was the arrival, during 2020, of a large cohort of new retail participants trading through mobile applications. The immediate effect was a surge in single-stock call buying, concentrated in short maturities and small orders.

The second-order effect is the interesting one. Retail buying of short-dated calls leaves dealers short those calls and therefore short gamma, in the language Chapter 18 introduces: they are obliged to buy the stock as it rises and sell as it falls, the opposite of the damping behavior in Example 10.6. Chapter 23 develops the mechanism in full. In heavily traded stocks during 2021, this hedging demand was large relative to the free float of the stock, and the feedback from option market to stock market became visible rather than theoretical.

A third change came from the exchange side. Once S&P 500 options expired every trading day, every session contained contracts with hours rather than weeks to run. These are **zero-days-to-expiration (0DTE)** options, and Figure 10.5 shows their share of index option volume rising from about 5% to more than half.

Why 0DTE options behave differently. An option with hours to live has almost no time value and enormous gamma, so a dealer hedging a book of them must trade the underlying frequently and in large amounts, carrying option trading into the index during the day. Customers both buy and sell them and the positions close within the day, so dealers carry no large net position

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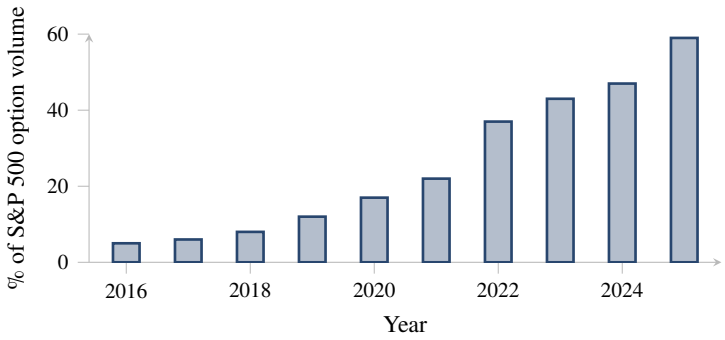


Figure 10.5: Share of S&P 500 index option volume expiring the same day. Approximate.

overnight. Dim, Eraker, and Vilkov (2024) find dealers' net 0DTE gamma small, though Brogaard, Han, and Won find that 0DTE trading raises volatility; Chapter 23 develops the mechanism and weighs that evidence.

Case Study 10.1 What retail bought, and what it cost

The retail options boom is usually discussed in terms of market stability. The more immediate question is what it did to the participants themselves.

Retail option trading is overwhelmingly *buying*, and heavily concentrated in short-dated, out-of-the-money calls: the cheapest contracts on the chain and, as Example 10.3 showed, those with the widest proportional spreads. A \$0.30 option with a \$0.05 spread costs over 16% of its value to enter and exit. A trader turning such a position over weekly faces a transaction cost burden that no directional skill plausibly overcomes.

Layered on top is a structural point from Part III. Buying options is buying volatility, and Chapter 19 documents that implied volatility exceeds subsequent realized volatility on average across most markets and periods. Persistent option buyers therefore pay a risk premium on top of the spread, and should know they are paying it.

None of this argues against retail participation. The cost is simply incurred where no statement shows it: in the spread and in the volatility premium.

SUMMARY

A call confers the right to buy at the strike and a put the right to sell; European options are exercisable only at expiration, American options at any time. A buyer's loss is capped at the premium and the buyer posts no margin; a writer's gain is capped at the premium and the writer's open-ended loss must be collateralized. This asymmetry, not the payoff formula, distinguishes the two sides in practice. An option's value is intrinsic value (what it would be worth exercised now) plus time value, which is largest near the money and decays to zero at expiration.

A standard US equity option covers 100 shares. Contracts are adjusted for splits, mergers, and special dividends to preserve aggregate value, but not for ordinary cash dividends, and an adjusted contract trades under a numbered root. The same right differs by underlying: index options settle in cash against a computed value, ETF options can be assigned early around dividends and are taxed differently, and an option on a futures contract delivers a margined position rather than an asset.

Reading an option chain reveals what no model supplies: where liquidity is, how large the proportional bid–ask spread is on cheap options, and whether the quotes are consistent

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with put–call parity. Large trades between professionals are quoted in implied volatility, so a volatility is a traded number and not a model output. On a hard-to-borrow stock the borrow fee appears as puts priced above parity, a financing cost rather than a view.

Naked writers post the greater of two formulas, recomputed daily and far exceeding the premium collected; covered and cash-secured writers post much less, and portfolio margining reduces requirements for hedged positions and thereby enables larger ones.

The OCC clears all US listed options and allocates exercise notices randomly among writers. Options in the money by one cent are exercised automatically at expiration, and positions struck near the closing price create pin risk: an unhedged position of unknown size discovered after the market has closed. The larger over-the-counter market holds most currency and interest rate options and all exotic options, and its buyer is exposed to the dealer's default unless the trade is collateralized or cleared.

Since 2020 zero-commission retail participation and index options expiring every trading day have reshaped the market, and same-day options now account for more than half of S&P 500 option volume. Their hedging carries option trading into the index during the day, and the retail cost lies in bid–ask spreads and the volatility risk premium rather than in commissions.

FURTHER READING

The OCC's *Characteristics and Risks of Standardized Options*, the disclosure document every US options customer receives, states the contract, exercise, and margin rules; the OCC's adjustment memoranda show the principle of Example 10.1 applied case by case, and exchange rulebooks give the current margin formulas. Bryzgalova, Pavlova, and Sikorskaya (2023) is the empirical study of retail option buying and payment for order flow behind Section 10.12's case study. Cboe's research note "The Rise of SPX 0DTE Options" documents the growth that Figure 10.5 plots, and Brogaard, Han, and Won's "Does 0DTE Options Trading Increase Volatility?" is the place to start on the open question Chapter 23 weighs. On payment for order flow, the brokers' Rule 606 routing reports, which disclose what each wholesaler pays for options orders, are the primary data.

PRACTICE QUESTIONS

10.1 An investor buys 5 call contracts at a quoted premium of \$2.85. What is the total cost, and what is the maximum loss?

10.2 Using Table 10.2, compute the proportional bid–ask spread for the 45-strike call and the 55-strike call. Explain why they differ and what it implies for strategy selection.

10.3 A stock trades at \$74 and pays no dividend. A trader writes 8 naked call contracts struck at \$85 for a premium of \$1.10. Compute the margin requirement from both formulas and state which binds.

10.4 A company declares a 2-for-1 stock split. State how the contract terms of an outstanding option are adjusted under each of the two conventions described in Section 10.2, and verify that the aggregate strike cost is preserved in both.

10.5 Explain why an option buyer posts no margin while a writer must, and relate the answer to Figure 10.1.

10.6 A trader is short 50 call contracts struck at \$30 and the stock closes at \$30.00 on expiration Friday. Describe the trader's position on Saturday morning and explain what could have been done on Friday to avoid it.

10.7 An option expiring in three hours and one expiring in three months are both at the money. Compare their intrinsic value, time value, and sensitivity to a \$1 move in the underlying.

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10.8 Distinguish a warrant from a listed call option on the same stock at the same strike. Which is worth more, and why?

10.9 An index option, a currency option, and an option on a futures contract differ in what changes hands on exercise. State what each delivers, and explain why exercising an option on a futures contract leaves the holder with a position to manage rather than an asset to keep.

10.10 Dealers quote many options in volatility rather than in price. Explain what the convention buys them, state what else must be agreed before a volatility quote determines a cash amount, and explain why such a quote stays meaningful while the underlying moves and a price quote does not.

10.11 An investor wants six months of downside protection on a $5 million S&P 500 portfolio, with the index at 5,250 and SPY at $525. Compare buying index puts (SPX) with buying SPY puts: the number of contracts, settlement and exercise, what each leaves the investor holding at expiration, and US tax treatment.

10.12 Name three distinct institution types that buy options and state, for each, the exposure it is buying or removing. For one of them, explain how its demand shows up in the prices the rest of the market sees.

FURTHER QUESTIONS

10.13 Using the 50-strike quotes in Table 10.2, repeat the put–call parity check of Example 10.3 at that strike. Does the chain remain consistent? Identify what you must assume about dividends for the two strikes to agree, and comment on whether that assumption is reasonable.

10.14 Section 10.12 argues that dealer hedging of short-dated retail call buying amplifies moves in the underlying, while Example 10.6 argues that dealer hedging near expiration damps them. Reconcile these. Precisely what determines the sign of the effect, and what would you need to observe to know which regime prevails today?

10.15 Two weeks before the covered call in Example 10.7 expires, the stock is at $55, the calls are quoted $2.50–$2.60, and tomorrow the stock goes ex-dividend with the $1.00 dividend of Example 10.3, after which the calls would be worth about $2.35. Explain why the investor may be assigned tonight, compute what an assignment costs against holding through the ex-dividend date, and state what the investor could have done today to avoid it.

10.16 Contract adjustments preserve aggregate value for splits and mergers but not for ordinary cash dividends. Explain the rationale for treating the two differently, and construct a case where the distinction transfers substantial value from one side of a contract to the other.

10.17 Same-day options do not accumulate an overnight position, which Section 10.12 offers as a reassuring feature. Construct the scenario in which this feature fails to protect the market, and identify what a regulator would need to monitor to see it developing.

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